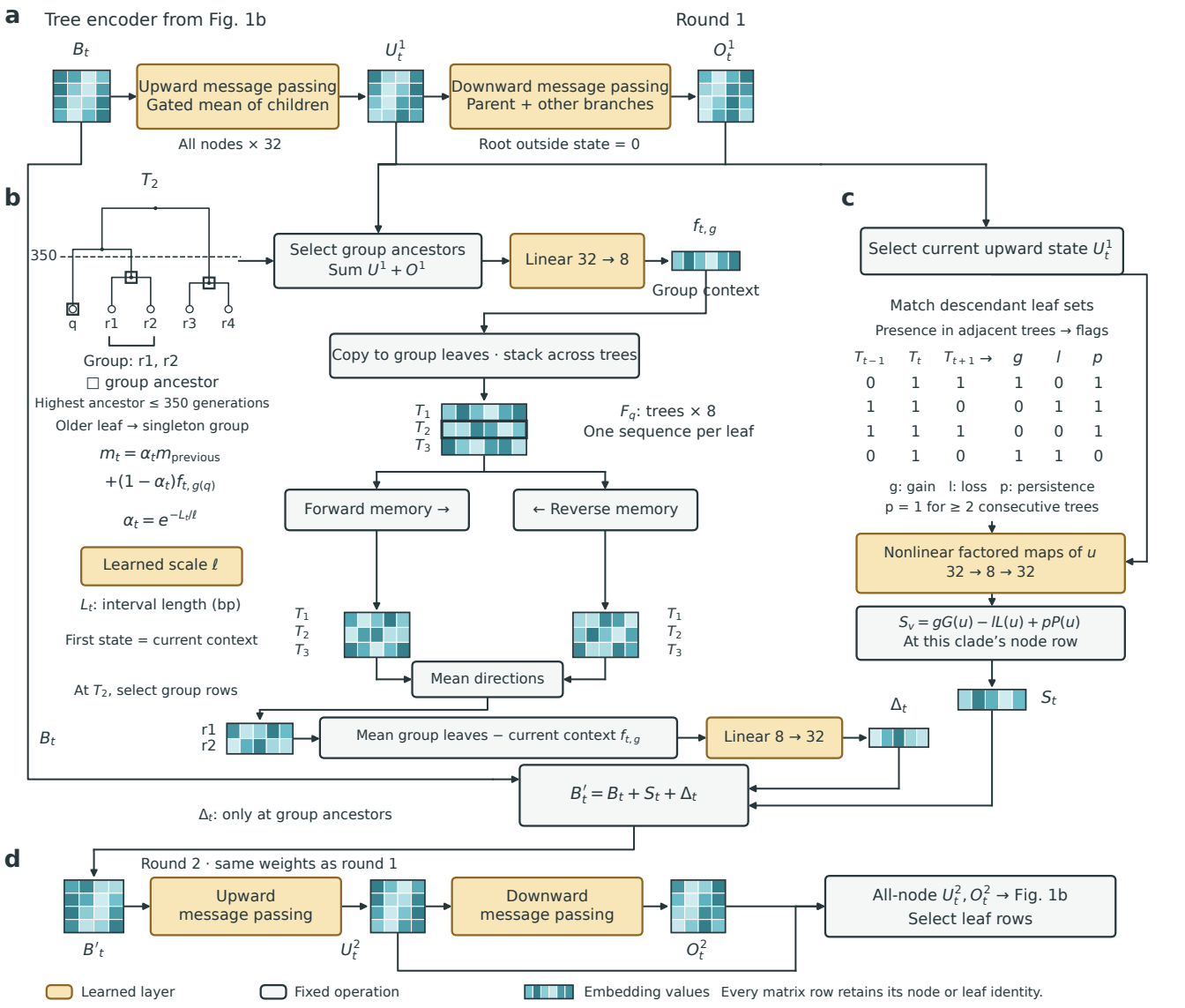
a
Tree encoder from Fig. 1b
Round 1
B_t
Upward message passing
Gated mean of children
All nodes × 32
U_t^1
Downward message passing
Parent + other branches
Root outside state = 0
O_t^1
b
T_2
350
q
r1
r2
r3
r4
Group: r1, r2
□ group ancestor
Highest ancestor ≤ 350 generations
Older leaf → singleton group
$m_t = \alpha_t m_{\text{previous}}$
$+(1-\alpha_t) f_{t,g(q)}$
$\alpha_t = e^{-L_t/\ell}$
Learned scale ℓ
L_t: interval length (bp)
First state = current context
At T_2, select group rows
Select group ancestors
Sum $U^1 + O^1$
Linear 32 → 8
$f_{t,g}$
Group context
Copy to group leaves · stack across trees
T_1
T_2
T_3
F_q: trees × 8
One sequence per leaf
Forward memory →
← Reverse memory
Mean directions
r1
r2
Mean group leaves − current context $f_{t,g}$
Linear 8 → 32
Δ_t
B_t
$B'_t = B_t + S_t + \Delta_t$
Δ_t: only at group ancestors
c
Select current upward state U_t^1
Match descendant leaf sets
Presence in adjacent trees → flags
T_{t-1} T_t T_{t+1} → g l p
0 1 1 1 0 1
1 1 0 0 1 1
1 1 1 0 0 1
0 1 0 1 1 0
g: gain l: loss p: persistence
p = 1 for ≥ 2 consecutive trees
Nonlinear factored maps of u
32 → 8 → 32
$S_v = gG(u) - lL(u) + pP(u)$
At this clade's node row
S_t
d
Round 2 · same weights as round 1
B'_t
Upward message passing
U_t^2
Downward message passing
O_t^2
All-node U_t^2, O_t^2 → Fig. 1b
Select leaf rows
Learned layer
Fixed operation
Embedding values
Every matrix row retains its node or leaf identity.